FINAL TECHNICAL DOCUMENTATION

Weather, Soil & Water Modeling Component

Smart Agricultural Water Management System — Rice, Egypt Nile Delta

Role	Weather, Soil & Water Modeling
Scope	Weather preprocessing, soil/AWC, rice parameters, ET_0, ETc, irrigation requirement and validation
Modeling seasons	2017–2023 documented rice seasons
Weather coverage	38 gridded locations, 2017-01-01 to 2026-10-25
Status	Core modeling pipeline completed; results are scenario-based and not field-validated

Core interpretation: The pipeline is reproducible and scientifically structured, but the final irrigation estimates are scenario estimates. The available literature benchmark does not overlap the modeled range, so the results must not be presented as measured farm water consumption or field-validated irrigation demand.

Contents

- 1. Executive Summary
- 2. Objective and Scope
- 3. End-to-End Pipeline and Data Lineage
- 4. Weather Data Analysis and Cleaning
- 5. Soil and Available Water Capacity Processing
- 6. Rice Crop Parameters and Reference Data
- 7. FAO-56 Reference Evapotranspiration (ET_0)
- 8. Rice Crop Evapotranspiration (ETc)
- 9. Irrigation Requirement Method
- 10. Validation and Benchmarking
- 11. Final Results Summary
- 12. Assumptions, Uncertainty and Limitations
- 13. Outputs for Team Integration
- 14. GitHub File Structure and Deliverables
- 15. Acceptance-Criteria Status
- 16. Conclusion
- Appendix A. Key Equations
- Appendix B. Selected Output Data Dictionary



Figure 1. Final modeling workflow implemented across notebooks 01–06.

1. Executive Summary

This component developed a reproducible weather–soil–crop water modeling pipeline for rice irrigation planning in the Egyptian Nile Delta. The workflow cleaned and prepared gridded NASA POWER weather data and mapped soil/AWC data, organized rice growth-stage and water-management parameters, calculated daily FAO-56 Penman–Monteith reference evapotranspiration (ET_0), estimated rice crop evapotranspiration (ETc), and built seasonal net and gross irrigation requirement scenarios with effective rainfall, paddy-rice water-management terms, soil sensitivity, and literature benchmarking.

Area	Final result
Weather	1,089,840 raw long-format records; 136,002 complete daily-point modeling rows across 38 locations.
Soil	38 mapped points prepared; 36 retained for modeling; 2 excluded as non-soil/water-body mapping cases.
ET_0	136,002 daily estimates; mean 4.868 mm/day; range 0.852–11.628 mm/day; no missing or negative ET_0.
ETc	39,900 rice-season daily rows and 266 location-season summaries; seasonal mean 1,115.7–1,141.1 mm.
Net irrigation	Main scenario mean 2,012.5–2,637.9 mm/season, equivalent to 8,452–11,079 m^3/feddan.
Validation	Literature 5th–95th percentile 678.6–1,570.5 mm/season; 0% overlap with main modeled range; caution required.

2. Objective and Scope

The assigned objective was to estimate rice water requirements using weather data, soil properties, crop growth stages, and documented agricultural water-management references, while explicitly separating calculated estimates from actual field measurements. The implemented scope covers the core scientific tasks requested in the project brief:

- Weather quality inspection, cleaning, unit verification and location/date preparation.
- Soil and available-water-capacity (AWC) inspection, quality screening and spatial matching.
- Rice crop coefficients (Kc), growth-stage durations and rice-specific water-management parameters.
- FAO-56 Penman–Monteith ET_0 calculation.
- $ETc = ET_0 \times Kc$ with stage-based Kc progression.
- Net irrigation requirement using effective rainfall, saturation, standing-water establishment, and seepage/percolation.
- Soil-adjusted seepage sensitivity and gross-irrigation efficiency sensitivity.
- Comparison against existing project scenarios and independent published experimental records.
- Water-depth-to-volume conversion using 1 mm over 1 feddan = 4.2 m^3.

3. End-to-End Pipeline and Data Lineage

Notebook	Purpose	Primary output
01_weather_data_analysis.ipynb	Inspect and clean NASA POWER weather data; create daily wide-format model input.	weather_daily_prepared.csv.gz; weather_daily_clean.csv.gz
02_soil_data_analysis.ipynb	Inspect soil/AWC, verify units, exclude invalid spatial points, preserve review flags.	soil_prepared.csv; soil_clean.csv
03_rice_crop_parameters.ipynb	Prepare crop calendar, Kc, stage durations, varieties and water-management parameters.	Six reference CSVs under data/reference/
04_et0_calculation.ipynb	Calculate daily FAO-56 ET_0.	eto_daily.csv.gz
05_etc_calculation.ipynb	Assign stages/Kc and calculate daily + seasonal rice ETc.	etc_daily.csv.gz; etc_seasonal.csv
06_irrigation_requirement_validation.ipynb	Calculate irrigation requirement, volume, sensitivities and validation comparisons.	irrigation_requirement_seasonal.csv; irrigation_validation_summary.csv; literature_irrigation_benchmark.csv

4. Weather Data Analysis and Cleaning

4.1 Raw data profile

Item	Result
Source	NASA POWER project weather table
Raw shape	1,089,840 rows × 6 columns
Locations	38 grid points
Date range	2017-01-01 to 2026-10-25
Parameters	8
Exact duplicate rows	0
Duplicate point-date-parameter keys	0
Missing values	1,292 total values, concentrated at the end of October 2026

4.2 Verified variables and units

Variable	Meaning	Unit
T2M	Daily mean air temperature	°C

T2M_MAX	Daily maximum air temperature	°C
T2M_MIN	Daily minimum air temperature	°C
RH2M	Relative humidity	%
PRECTOTCORR	Corrected precipitation	mm/day
WS2M	Wind speed at 2 m	m/s
PS	Surface pressure	kPa
ALLSKY_SFC_SW_DWN	All-sky surface shortwave radiation	MJ/m²/day

4.3 Cleaning decisions

- Converted obs_date to datetime and pivoted long-format weather records to one row per point and date.
- Merged the 38-point grid table using point_id and retained latitude, longitude, district_id and the label “gridded estimate”.
- Required complete ET_0 inputs and rainfall for the modeling dataset. Prepared rows = 136,230; clean modeling rows = 136,002; 228 rows excluded.
- Physical checks found 0 RH values outside 0–100%, 0 negative rainfall values, 0 negative wind speeds, 0 cases with Tmin > Tmax, and 0 negative solar-radiation values.
- IQR outliers were reviewed rather than automatically deleted because statistical extremeness does not imply a physically invalid observation.
- Precipitation above the 99.9th-percentile threshold (64.02 mm/day) was flagged for review rather than removed: 135 rows flagged.
- Weather values are gridded estimates and must not be described as measurements from individual farms.

5. Soil and Available Water Capacity Processing

Item	Result
Soil table	38 rows × 15 columns
AWC table	38 rows × 9 columns
Grid table	38 rows × 5 columns
Exact duplicates	0 in soil and AWC
Complete soil records	36
Excluded points	grid_0002 and grid_0028
Prepared rows	38
Clean modeling rows	36

5.1 Data quality and exclusions

Two mapped points lacked valid soil properties and were excluded from soil-based modeling: grid_0002 was screened as a non-soil mapping case and grid_0028 as a water-body/no-data case. Their AWC values of zero were not interpreted as real zero water-storage capacity.

AWC use recommendation	Count
REVIEW_BEFORE_MODEL_USE	19
CANDIDATE_AFTER_SPATIAL_CHECK	17
EXCLUDE	2

5.2 Clean-soil summary

Property	Mean	Range	Unit
Sand	39.67	21–50	% by weight
Silt	28.83	24–40	% by weight
Clay	31.50	20–55	% by weight
Bulk density	1.467	1.33–1.64	g/cm³
Organic carbon	0.715	0.62–0.901	% by weight
pH water	8.053	7.3–8.5	unitless
AWC (SMU)	123.75	89–172	mm/m
Rootable AWC estimate	125.11	89–178	mm

Interpretation: Several texture/organic-carbon/pH groups were IQR outliers, but they appeared as coherent mapped soil groups, not random errors. They were retained. All remaining AWC values are provisional/not field validated.

6. Rice Crop Parameters and Reference Data

The crop-parameter notebook preserved the distinction between documented source values, ranges, and intentionally missing values. General rice parameters were not falsely assigned to specific varieties.

Parameter / stage	Value	Status / use
Initial Kc	1.05	General rice reference; not locally calibrated
Development Kc	No single fixed source value	Interpolated linearly from 1.05 to 1.20 during ETc calculation
Mid-season Kc	1.20	General rice reference; not locally calibrated
Late-season Kc	0.60–0.90	Range retained as low/high scenario
Initial stage length	30 days	Reference
Development stage length	30 days	Reference
Mid-season stage length	60 days	Reference
Late-season stage length	30 days	Reference
Reference growing season	150 days	Sum of four stages
Soil saturation requirement	200 mm	Literature assumption requiring local validation
Seepage/percolation	4–8 mm/day	Literature range requiring local validation
Water-layer establishment	100 mm	Literature assumption requiring local validation

- Calendar/season records cover 2017–2023 only; the pipeline did not silently extend the rice calendar into 2024–2026.
- 19 variety records were merged with 18 detailed variety-reference rows. NOT_SPECIFIED is a placeholder rather than a real named variety.
- Sakha Super 300 lacks a season-length value and was kept missing rather than imputed.

7. FAO-56 Reference Evapotranspiration (ET_0)

Daily ET_0 was calculated with the FAO-56 Penman–Monteith method using the cleaned temperature, humidity, wind, pressure and solar-radiation inputs. Mean relative humidity was used to estimate actual vapor pressure because

RHmax/RHmin were not available. Daily soil heat flux was taken as zero; clear-sky radiation used the near-sea-level approximation employed in the notebook.

$$ET_0 = [0.408\,\Delta\,(R_n - G) + \gamma\,(900 / (T + 273))\,u_2\,(e\square - e_a)] / [\Delta + \gamma\,(1 + 0.34\,u_2)]$$

Daily FAO-56 Penman–Monteith formulation; output in mm/day.

Metric	Result
Rows	136,002
Mean ET_0	4.868 mm/day
Standard deviation	1.987 mm/day
Minimum	0.852 mm/day
Median	4.797 mm/day
Maximum	11.628 mm/day
Missing ET_0	0
Negative ET_0	0
Radiation ratio > 1 flags	0

8. Rice Crop Evapotranspiration (ETc)

$$ETc = ET_0 \times Kc$$

Only the documented 2017–2023 rice seasons were modeled. The 150-day reference season was divided into 30/30/60/30-day stages. Development Kc increased linearly from 1.05 to 1.20. Late-season Kc decreased linearly from 1.20 to either 0.60 (low) or 0.90 (high), preserving the documented endpoint range.

Metric	Low scenario	High scenario
Daily rice-season rows	39,900	39,900
Mean Kc	1.0935	1.1245
Kc range	0.60–1.20	0.90–1.20
Mean daily ETc	7.438 mm/day	7.607 mm/day
Daily ETc range	2.398–12.209	3.446–12.209
Seasonal records	266	266
Mean seasonal ETc	1,115.7 mm	1,141.1 mm
Seasonal ETc range	1,012.3–1,234.9 mm	1,036.6–1,261.4 mm
Missing / negative ETc	0 / 0	0 / 0

9. Irrigation Requirement Method

ETc alone was not treated as the irrigation requirement. The seasonal paddy-rice water balance included effective rainfall and rice-specific management/loss terms.

$$\text{Net Irrigation} = ETc + SAT + PERC + WL - Pe$$

Term	Implementation
ETc	Low/high seasonal ETc from Notebook 05
SAT	200 mm soil-saturation / land-preparation requirement

PERC	Main scenario: 4–8 mm/day × 150 season days = 600–1,200 mm
WL	100 mm establishment/standing-water layer
Pe	Monthly effective rainfall converted back to daily contributions
Stored soil water	0 mm adjustment; actual initial soil moisture unavailable
Water-volume conversion	Volume (m³) = depth (mm) × area (feddans) × 4.2

9.1 Effective rainfall

Pe = 0.8P − 25 when P > 75 mm/month; otherwise Pe = 0.6P − 10

Negative effective rainfall was clipped to zero. Monthly Pe was allocated to rainy days in proportion to observed daily rainfall.

9.2 Main net irrigation results

Metric	Low	High
Mean depth	2,012.5 mm/season	2,637.9 mm/season
Depth range	1,912.3–2,134.9 mm	2,536.6–2,761.4 mm
Mean volume	8,452.4 m³/feddan	11,079.2 m³/feddan
Volume range	8,031.8–8,966.6 m³/feddan	10,653.7–11,597.8 m³/feddan

9.3 Soil-adjusted seepage sensitivity

Clay condition	Seepage sensitivity
Clay ≥ 40%	1–3 mm/day
25% ≤ clay < 40%	2–5 mm/day
Clay < 25%	4–8 mm/day
No usable soil data	Default 4–8 mm/day

Across the 266 location-season rows, the soil sensitivity produced 126 clay<25% rows, 70 clay 25–40% rows, 56 clay≥40% rows, and 14 rows without usable soil (2 excluded grid points × 7 seasons).

Metric	Low	High
Mean soil-adjusted depth	1,838.8 mm	2,361.6 mm
Range	1,612.4–2,134.8 mm	1,946.8–2,761.3 mm
Mean volume	7,722.9 m³/feddan	9,918.7 m³/feddan
Volume range	6,772.0–8,966.3 m³/feddan	8,176.7–11,597.4 m³/feddan

9.4 Gross irrigation efficiency sensitivity

No site-specific measured irrigation efficiency was available. Therefore, gross water supply was reported only as a sensitivity analysis, not as a measured value.

Gross irrigation = Net irrigation / application efficiency

Efficiency	Mean gross low	Mean gross high
60%	14,087 m³/feddan	18,465 m³/feddan
70%	12,075 m³/feddan	15,827 m³/feddan
80%	10,565 m³/feddan	13,849 m³/feddan

10. Validation and Benchmarking

10.1 Existing project scenarios

The existing water_scenario.csv table contained 266 records. Every record was explicitly labeled NOT_MEASURED and SCENARIO_ESTIMATE_NOT_FIELD_VALIDATED. Accordingly, it was used only for internal consistency comparison, not as validation ground truth.

Comparison	Mean difference (new − old)
Low irrigation depth	−205.44 mm
High irrigation depth	−280.01 mm
Low volume	−862.84 m³/feddan
High volume	−1,176.05 m³/feddan
Stage-based ETc vs previous ETc	−119.32 mm/season

10.2 Independent literature benchmark

Water-related literature records were filtered to seasonal/total irrigation metrics. Sixty-five records were retained and all 65 were converted to a common depth unit (mm/season); no records remained unconverted.

Literature benchmark statistic	Irrigation depth
Minimum	640.5 mm
5th percentile	678.6 mm
Median	1,176.1 mm
Mean	1,162.1 mm
95th percentile	1,570.5 mm
Maximum	1,614.3 mm

> **Critical validation result:** The main modeled irrigation range (1,912.3–2,761.4 mm/season) did not overlap the literature 5th–95th percentile benchmark (678.6–1,570.5 mm/season). The overlap rate was 0% across all 266 modeled location-season rows. This is a discrepancy that requires caution, not a result to hide or relabel as validation success.

Plausible contributors documented in the notebook include different irrigation regimes and experimental treatments, differences in seepage/percolation and soil conditions, and the model's explicit inclusion of land preparation and standing-water requirements. The model must therefore be described as a scenario-based water-requirement framework rather than field-validated irrigation demand.

11. Final Results Summary

Stage	Key final result	Interpretation
Weather	136,002 complete daily-point rows	Suitable input for ET_0; gridded, not farm measured.
Soil	36 usable points / 38 total	2 excluded; remaining AWC provisional.
ET_0	0.852–11.628 mm/day; mean 4.868	No missing/negative outputs.
ETc	1,012–1,261 mm/season across low/high endpoints	150-day rice reference season, 2017–2023.
Main net irrigation	1,912–2,761 mm/season	Scenario estimate using 4–8 mm/day seepage.
Main field volume	8,032–11,598 m³/feddan	Net field requirement range.

Soil sensitivity	1,612–2,761 mm/season	Secondary sensitivity, not field-calibrated.
Literature benchmark	679–1,571 mm/season (5th–95th)	No overlap; validation caution required.
Gross supply	Efficiency sensitivity only	60/70/80%; no measured efficiency.

12. Assumptions, Uncertainty and Limitations

- NASA POWER observations are gridded estimates, not measurements from individual rice farms.
- The weather record extends into 2026, but the rice season/calendar reference is documented only for 2017–2023; ETc and irrigation modeling therefore use those seven seasons.
- Mean RH was used for vapor-pressure estimation because daily RHmax/RHmin were not available.
- Clear-sky radiation used a near-sea-level approximation rather than point-specific elevation.
- Soil/AWC data are mapped/provisional and not laboratory measurements from the modeled farms. Two points lacked usable soil data.
- Rice Kc and stage durations are general reference values and are not locally calibrated for every Egyptian variety/site.
- Saturation (200 mm), water-layer establishment (100 mm) and main seepage/percolation (4–8 mm/day) are literature/reference assumptions requiring local validation.
- Available water capacity was not subtracted as actual stored water because initial soil moisture was not measured.
- Irrigation efficiency was not measured; 60%, 70% and 80% gross-supply calculations are sensitivity scenarios only.
- Existing project water scenarios are estimates and explicitly not field measured.
- The independent literature benchmark contains varying sites, treatments and irrigation methods; it is useful as a benchmark but not a perfect one-to-one field validation set.
- The 0% benchmark overlap indicates substantial model–literature discrepancy. Field observations of applied irrigation, percolation, initial soil moisture, water-layer management and application efficiency are needed before claiming field-level accuracy.

13. Outputs for Team Integration

13.1 Satellite / image-processing team

- Join by point_id and obs_date/season_year.
- Use latitude/longitude and growth_stage for matching weather/water estimates to NDVI/EVI or other satellite indices.
- Treat modeled water variables as estimates, not measured irrigation labels.

13.2 Data cleaning and ML/DL team

- Use weather_daily_clean.csv.gz and soil_clean.csv as documented cleaned features.
- Use ET_0/ETc/irrigation outputs only with the status and uncertainty fields preserved.
- Do not use water_scenario values or the final irrigation estimates as ground-truth labels without an explicit modeling objective and suitable validation.

13.3 Optimization / chatbot / web application team

- Use the net irrigation range as a scenario output with units and assumptions shown to users.
- Convert depth to volume using depth × feddans × 4.2.
- If presenting gross supply, require the irrigation-efficiency assumption as an explicit input or scenario selector.
- Expose warnings when no local soil measurement, initial soil moisture, measured efficiency, or field validation is available.

14. GitHub File Structure and Deliverables

Path	Status / purpose
weather_soil_water_modeling/notebooks/01_weather_data_analysis.ipynb	Complete — weather preprocessing and quality checks

weather_soil_water_modeling/notebooks/02_soil_data_analysis.ipynb	Complete — soil/AWC processing
weather_soil_water_modeling/notebooks/03_rice_crop_parameters.ipynb	Complete — crop reference preparation
weather_soil_water_modeling/notebooks/04_et0_calculation.ipynb	Complete — daily ET_0
weather_soil_water_modeling/notebooks/05_etc_calculation.ipynb	Complete — daily/seasonal ETc
weather_soil_water_modeling/notebooks/06_irrigation_requirement_validation.ipynb	Complete — irrigation requirement and validation

14.1 Processed outputs

- weather_soil_water_modeling/data/processed/weather_daily_prepared.csv.gz
- weather_soil_water_modeling/data/processed/weather_daily_clean.csv.gz
- weather_soil_water_modeling/data/processed/soil_prepared.csv
- weather_soil_water_modeling/data/processed/soil_clean.csv
- weather_soil_water_modeling/data/processed/eto_daily.csv.gz
- weather_soil_water_modeling/data/processed/etc_daily.csv.gz
- weather_soil_water_modeling/data/processed/etc_seasonal.csv
- weather_soil_water_modeling/data/processed/irrigation_requirement_seasonal.csv
- weather_soil_water_modeling/data/processed/irrigation_validation_summary.csv
- weather_soil_water_modeling/data/processed/literature_irrigation_benchmark.csv

14.2 Reference outputs

- weather_soil_water_modeling/data/reference/crop_calendar_reference.csv
- weather_soil_water_modeling/data/reference/crop_water_parameters_reference.csv
- weather_soil_water_modeling/data/reference/season_reference.csv
- weather_soil_water_modeling/data/reference/rice_variety_reference_prepared.csv
- weather_soil_water_modeling/data/reference/rice_growth_stage_kc_reference.csv
- weather_soil_water_modeling/data/reference/rice_parameter_sources.csv

15. Acceptance-Criteria Status

Acceptance item	Status	Evidence in pipeline
Weather and soil inspected; units documented	Complete	Notebooks 01–02 and prepared datasets
Reproducible preprocessing	Complete	Notebook cells and version-controlled outputs
Rice stages/Kc traceable	Complete	Notebook 03 reference tables
ET_0 calculated and checked	Complete	Notebook 04; 136,002 valid outputs
ETc calculated consistently	Complete	Notebook 05; 39,900 daily / 266 seasonal rows
Irrigation method scientifically structured	Complete with limitations	Notebook 06 main + sensitivity scenarios
Water depth converted to volume	Complete	mm × feddans × 4.2
Existing scenarios/experimental records reviewed	Complete	Scenario tables + 65 literature irrigation records
Results compared with references	Complete; discrepancy found	0% overlap correctly documented
Limitations/assumptions documented	Complete	Notebook 06 and this report
Outputs ready for team integration	Complete for core modeling outputs	Processed/reference datasets with join keys

Remaining repository packaging work: The core modeling work is complete. The project brief also recommends reusable src/ modules and tests. Those can be extracted from the completed notebooks without changing the scientific results documented here.

16. Conclusion

The Weather, Soil & Water Modeling component now provides a complete, reproducible modeling chain from raw gridded weather and mapped soil data to daily ET_0, stage-based rice ETc, seasonal irrigation requirement scenarios, water volumes, efficiency sensitivities, and literature comparison. The implementation satisfies the central scientific objective of separating crop evapotranspiration from total paddy-rice water requirement and explicitly labeling assumptions where direct measurements are unavailable.

> **Final conclusion:** The calculated irrigation values are scientifically structured scenario estimates, not verified farm irrigation demand. The independent literature benchmark is substantially lower and shows 0% overlap with the main modeled range. Any farmer-facing or optimization use should therefore preserve uncertainty warnings and prioritize local field measurements for calibration before operational deployment.

Appendix A. Key Equations

$$ETc = ET_0 \times Kc$$

$$\text{Net irrigation} = ETc + SAT + PERC + WL - Pe$$

$$\text{Gross irrigation} = \text{Net irrigation} / \text{irrigation efficiency}$$

$$\text{Water volume } (m^3) = \text{Water depth (mm)} \times \text{Area (feddans)} \times 4.2$$

$$Pe = 0.8P - 25 \text{ for } P > 75 \text{ mm/month; } Pe = 0.6P - 10 \text{ otherwise}$$

Appendix B. Selected Output Data Dictionary

Field	Meaning	Unit / note
point_id	Spatial grid identifier	Join key
obs_date	Daily observation/model date	Date
season_year	Documented rice season year	2017–2023
ET0_mm_day	Reference evapotranspiration	mm/day
growth_stage	Rice growth stage	initial/development/mid_season/late_season
kc_low / kc_high	Scenario crop coefficient	dimensionless
ETc_low_mm_day / ETc_high_mm_day	Daily crop evapotranspiration	mm/day
ETc_low_mm / ETc_high_mm	Seasonal crop evapotranspiration	mm/season
effective_rain_mm	Estimated effective rainfall over modeled season	mm
net_irrigation_low_mm / high_mm	Main seasonal field irrigation scenario	mm/season
net_low_m3_per_feddan / high	Main field water-volume scenario	m^3/feddan
soil_adjusted_low_mm / high_mm	Soil seepage sensitivity result	mm/season
gross_*_eff_60/70/80	Gross supply sensitivity	mm or m^3/feddan; assumed efficiency
literature_range_overlap	Whether main modeled interval overlaps literature 5–95% range	Boolean
actual_irrigation_status	Field-measurement status label	NOT_FIELD_MEASURED
main_model_status	Model status	FAO_BASED_SCENARIO_NOT_FIELD_VALIDATED

Source Basis for This Documentation

This documentation was compiled directly from the completed project notebooks 01–06, their recorded outputs, generated processed/reference datasets, and the Team Member 2 task brief. Numerical results are reported as produced by those notebooks. No field accuracy is claimed beyond what the source materials support.